

Vectors: Components and Operations

ANSWER KEY



Vector arithmetic

- 1 Let $\mathbf{u} = \langle 3, -2 \rangle$ and $\mathbf{v} = \langle -1, 4 \rangle$. Find:

a $\mathbf{u} + \mathbf{v} = \langle 2, 2 \rangle$

b $3\mathbf{u} - 2\mathbf{v} = \langle 11, -14 \rangle$

c $|\mathbf{u}| = \sqrt{13}$

- 2 Find the component form and magnitude of the vector from $P(1, 3)$ to $Q(5, -2)$.

$\vec{PQ} = \langle 4, -5 \rangle$, $|\vec{PQ}| = \sqrt{16 + 25} = \sqrt{41}$.

- 3 Find the unit vector in the direction of $\mathbf{v} = \langle 6, -8 \rangle$.

$|\mathbf{v}| = 10$, so $\hat{\mathbf{v}} = \langle \frac{3}{5}, -\frac{4}{5} \rangle$.

- 4 Write the vector of magnitude 12 making an angle of 150° with the positive x-axis in component form (exact values).

$\langle 12\cos 150^\circ, 12\sin 150^\circ \rangle = \langle -6\sqrt{3}, 6 \rangle$.

Applications

- 5 A plane flies due east at 200 mph while the wind blows due north at 50 mph.

a Find the magnitude of the plane's resultant velocity, to one decimal place.

$\sqrt{200^2 + 50^2} = \sqrt{42500} \approx 206.2$ mph.

b Find the direction of the resultant, as an angle north of east to two decimal places.

$\theta = \tan^{-1}\left(\frac{50}{200}\right) \approx 14.04^\circ$ north of east.

- 6 Two forces $\mathbf{F}_1 = \langle 30, 0 \rangle$ N and $\mathbf{F}_2 = \langle 0, 40 \rangle$ N act on an object. Find the magnitude of the resultant force.

$|\mathbf{F}_1 + \mathbf{F}_2| = |\langle 30, 40 \rangle| = 50$ N.